

Removing the workspace layers of Llama-3.1-8B

Where does a language model *think*? Finding and removing the “workspace” layers of Llama-3.1-8B

A hands-on interpretability walkthrough: we watch concepts form layer-by-layer inside Llama-3.1-8B-Instruct, quantify which layers carry meaning, then delete layers — one at a time and in groups — and measure exactly how generation degrades. Then we connect it to Anthropic’s 2026 “global workspace” result. Everything runs on a single T4 GPU.

TL;DR. Topic information becomes **decodable within the first few layers and stays decodable everywhere** ($kNN \approx 0.9$ from layer 2 on). But through the first ~half of the network the classes are **visually entangled** — they collapse onto overlapping manifolds (silhouette ≈ 0) — and only **re-separate into clean clusters in the second half**. That entangled-but-decodable middle is where meaning is *worked on*. Deleting **one** middle layer barely dents generation; deleting the **whole band** collapses reasoning while leaving shallow recall intact — the same asymmetry Anthropic report when they ablate the **J-space / global workspace** [10]. Across **100 queries** (§6b), removing the band breaks **creative/abstract generation first and single-step recall last**.

0. Setup

- **Model:** meta-llama/Llama-3.1-8B-Instruct — 32 residual decoder layers, hidden 4096.
- **Compute:** one NVIDIA T4 (16 GB), model in 4-bit (nf4).

- **Two hooks:** `output_hidden_states=True` gives **33** residual-stream tensors (L_0 = embeddings; L_k = **output of decoder layer $k-1$** — mind this off-by-one), and a **layer-ablation wrapper** that turns any decoder block into an identity so the residual stream skips it (§4).
- **Data:** 400 texts, 4 balanced topic classes from 20-Newsgroups (*space*, *baseball*, *graphics*, *mideast*), mean-pooled per layer.

Reproduce from this repo — see §8.

1. First: KL divergence is not entropy

The question that started this. They’re related but distinct:

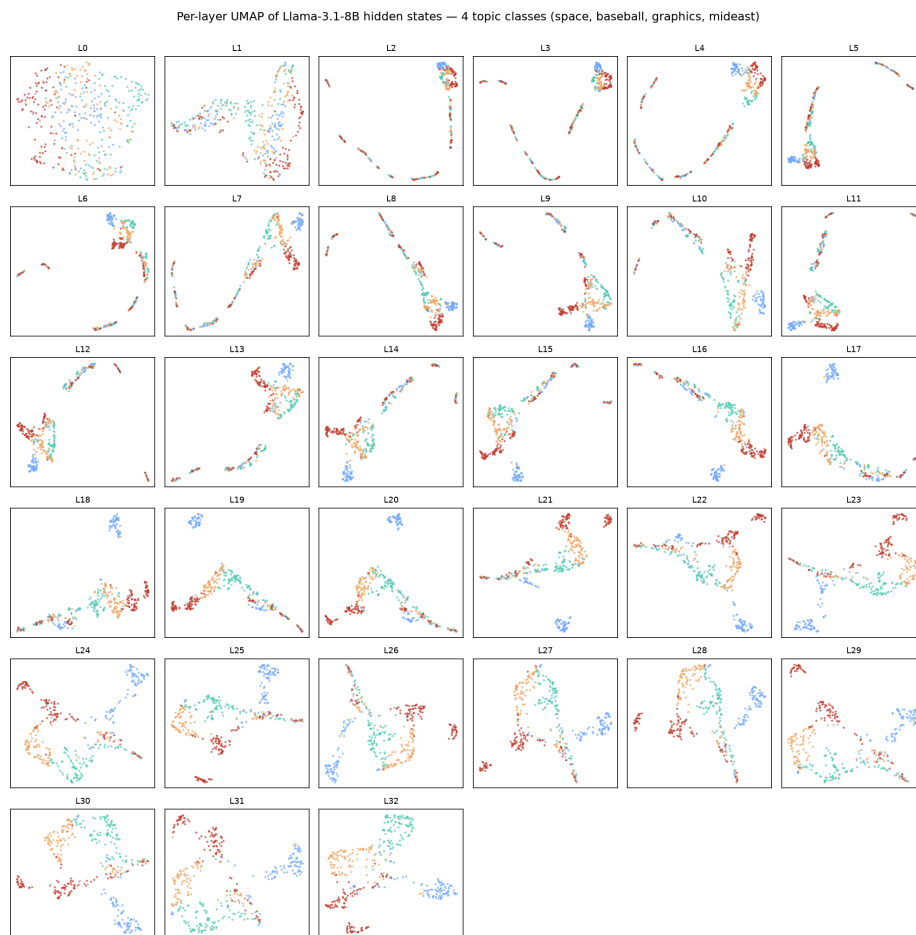
Quantity	Formula	Measures	# distributions
Entropy $H(p)$	$-\sum p \cdot \log p$	uncertainty of <i>one</i> distribution	1
Cross-entropy $H(p, q)$	$-\sum p \cdot \log q \rightarrow$ perplexity = $\exp(H)$	coding p with q	2
KL divergence $D_{KL}(p q)$	$\sum p \cdot \log(p/q) =$ $H(p, q) - H(p)$	how far p is from reference q	2

- **Entropy** is about *one* distribution — how spread/uncertain it is.
- **KL divergence** (“relative entropy”) is about *two* — how far the ablated model’s next-token distribution p_{ablated} moved from the original p_{baseline} . It is ≥ 0 , $= 0$ iff identical, and **not symmetric**.

So “measure the KL divergence of removing a layer” = compute $D_{KL}(p_{\text{ablated}} || p_{\text{baseline}})$ at each position — *how much did deleting this layer change the predictions?* We also report the **entropy** $H(p_{\text{ablated}})$ separately, because a broken model can move two ways: become **uncertain** (entropy $\uparrow$, distribution flattens) or get **stuck** (entropy $\downarrow$, repeats). KL catches both; entropy tells you which.

2. Experiment 1 — watch the representation reorganise, layer by layer

Push the 4-class set through the model, mean-pool the hidden state at each layer, project each layer to 2D with UMAP [8], colour by class:



Per-layer UMAP grid

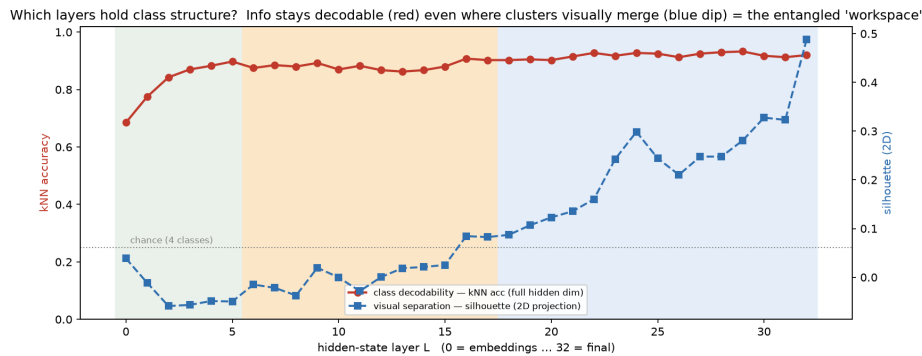
- **Embeddings / early ($\approx$ L0–L5).** L0 is a diffuse cloud; by L2–L6 the points collapse onto thin, folded **1-D manifolds** with the four topics **interleaved** along the curve.
- **Middle ($\approx$ L6–L16).** Still entangled — clusters overlap heavily; one class (graphics, blue) starts to peel off around L11–L14.
- **Late ($\approx$ L18–L32).** Clean **4-way separation** emerges and sharpens all the way to the final layer.

The representation goes **entangle** $\rightarrow$ **re-specialise**, echoing the logit lens [1], the tuned lens [2], and Tenney et al.’s “BERT rediscovers the classical NLP pipeline” [4].

Careful: UMAP/t-SNE are nonlinear 2-D projections [8]. “Looks entangled” is a statement about the *projection*, not about whether the class is *present*. Which is what §3 checks — and the answer flips the naive reading.

3. Experiment 1b — decodable vs. visually separated (this gap is the workspace)

For every layer, two measurements on the same embeddings: **kNN accuracy on the full 4096-d state** (can a classifier still read the class? = *decodability*) and **silhouette on the 2-D projection** (how visually separated? = *apparent separation*).



Separability across layers

- **Decodability (red) is high everywhere:** 0.68 at embeddings → ~0.90 by layer 5, and it stays 0.86–0.93 for the entire rest of the network. The class is *always* linearly-ish readable from layer 2 onward.
- **Visual separation (blue) tells the opposite story early:** silhouette is ≈ 0 (even negative) through roughly L1–L15, then climbs steeply to 0.49 by the final layer.

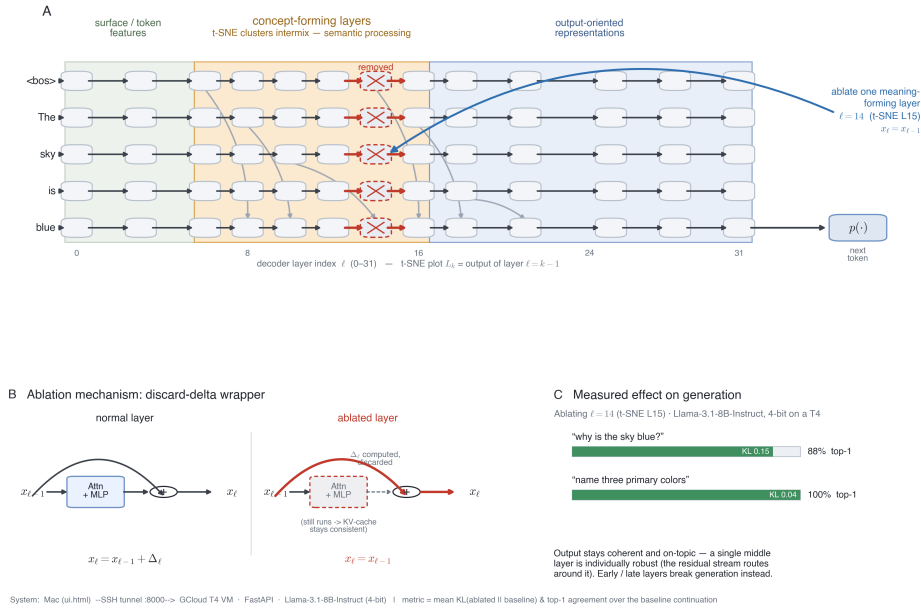
The **gap** — high decodability, near-zero visual separation — is the signature of a **distributed, entangled** code: the information didn't leave, it went into overlapping directions (**superposition** [5]). This is exactly the regime Anthropic's workspace lives in: their J-space is only ~6–10% of **activation variance** yet carries the concepts the model reasons over [10].

4. Experiment 2 — turning a layer off

Every decoder block is residual: $x_{out} = x_{in} + \text{attn}(x_{in}) + \text{mlp}(x_{in})$. To *remove* layer ℓ we wrap it so it still runs (keeping the KV-cache consistent) but returns its **input** unchanged, so the residual stream skips it: $x_{out} = x_{in}$. Reversible, no retraining — the layer-level analogue of activation patching / causal tracing [3].

Removing a meaning-forming layer from Llama-3.1-8B-Instruct

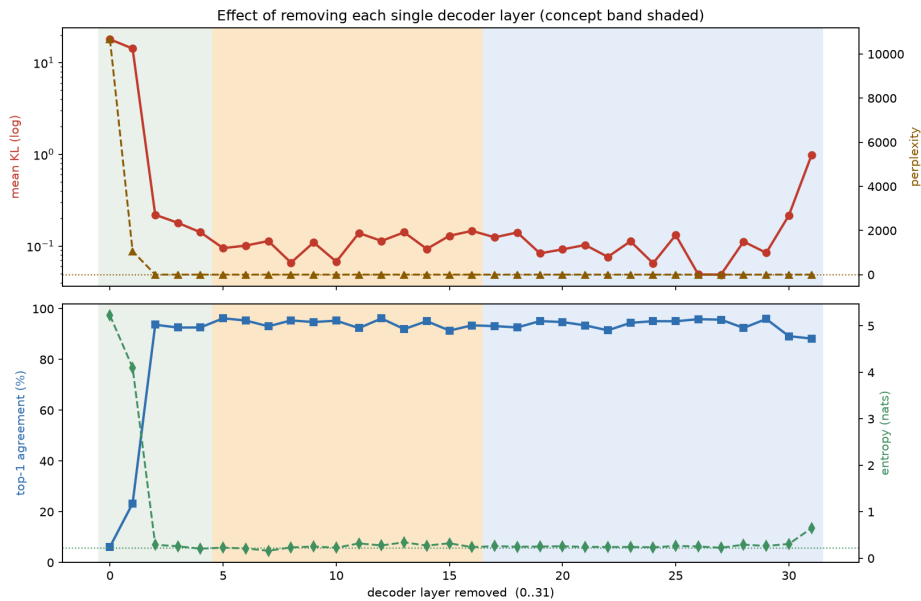
Ablate one middle decoder layer (residual stream passes straight through), regenerate, and measure the shift.



Architecture: removing a layer

5. Experiment 3 — remove each layer, one at a time

For 8 prompts we generate a greedy baseline continuation, then — teacher-forced on it — re-run with one layer ablated and measure KL, top-1 agreement, entropy, perplexity:



Single-layer ablation sweep

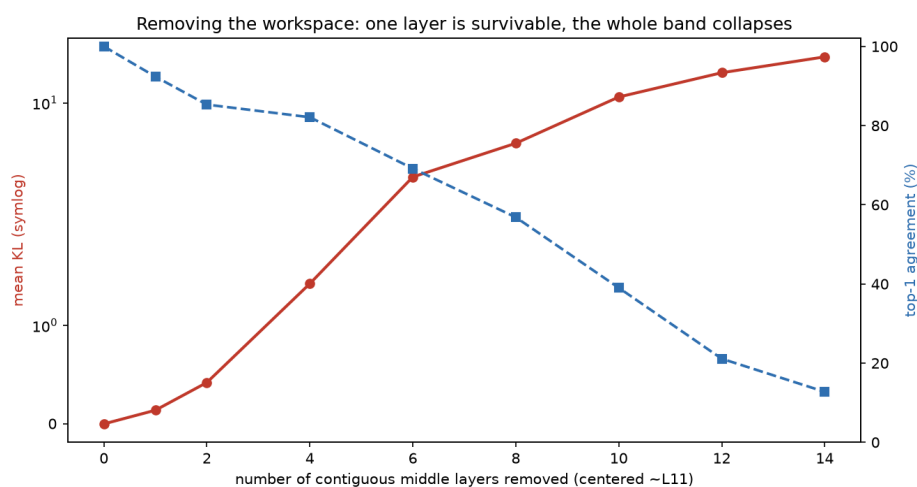
Not a symmetric U — it's an **early cliff**:

- **Layers 0–1 are load-bearing.** Remove L0 → KL ≈ 18 , top-1 6%, perplexity $\sim 10,700$. Remove L1 → KL ≈ 14 , top-1 23%. The model is destroyed.
- **Every other single layer is remarkably survivable.** L2–L29: KL ≈ 0.05 – 0.22 , top-1 89–96%, perplexity barely above the 1.10 baseline. Middle layers are individually the *most redundant* — which is exactly why depth-pruning methods delete them [6, 7].
- **The final layer matters a bit more:** L31 → KL ≈ 0.98 , top-1 88%.

One missing middle block? The residual stream just routes around it.

6. Experiment 4 — remove the whole band → the workspace collapses

Individually redundant $\neq$ collectively expendable. Delete a **growing contiguous window** of middle layers (centred $\sim$ L11) and watch the model fall over:



Cumulative middle-band removal

layers removed	KL	top-1	perplexity
1	0.14	92%	1.2
2	0.42	85%	1.4
4	1.4	82%	1.7
6	3.8	69%	2.8
8	5.9	57%	7.0
10	10.8	39%	33
12	14.8	21%	183

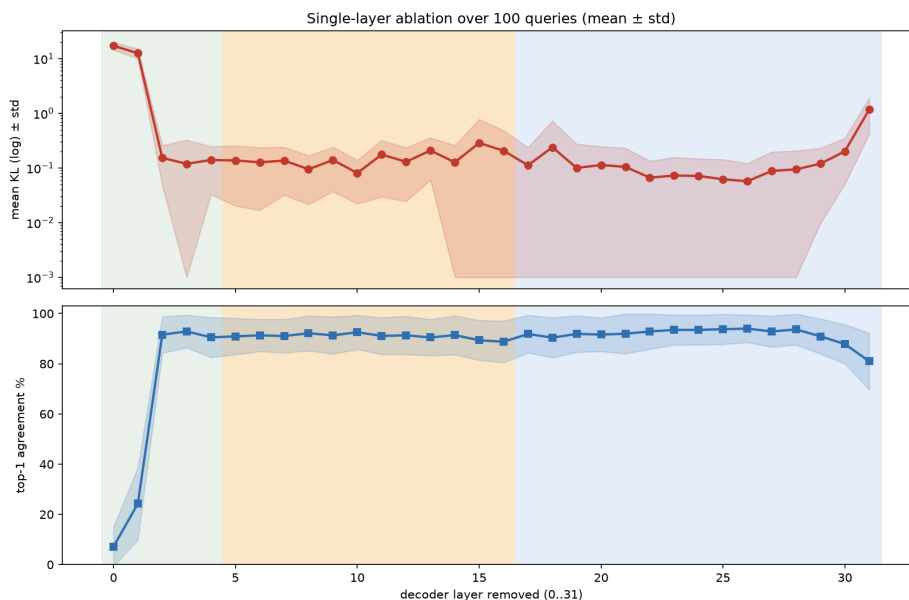
layers removed	KL	top-1	perplexity
14	18.2	13%	2658

A graceful-then-catastrophic collapse. (An independent interactive run agrees: removing the whole 12-layer concept band on “*why is the sky blue?*” gave $KL \approx 15.6$, top-1 20%, and the output degenerated into “*...the sky is a beautiful blue and the sky is a beautiful blue. It’s a blue collar, a blue collar...*” — the **word** “blue” survived, the **explanation** did not.)

6b. Experiment 5 — scaling to 100 queries, and which capability breaks first

Experiments 3–4 used 8 prompts. Here we repeat both on **100 queries**, balanced across five capability types (20 each: **recall**, **reasoning**, **math**, **translation**, **creative**), and add the question the workspace paper really cares about: *when you remove the band, does **reasoning** break before **recall**?* (Raw per-query numbers: [per_query_results.csv / .json](#); aggregates in [metrics_100.json](#).)

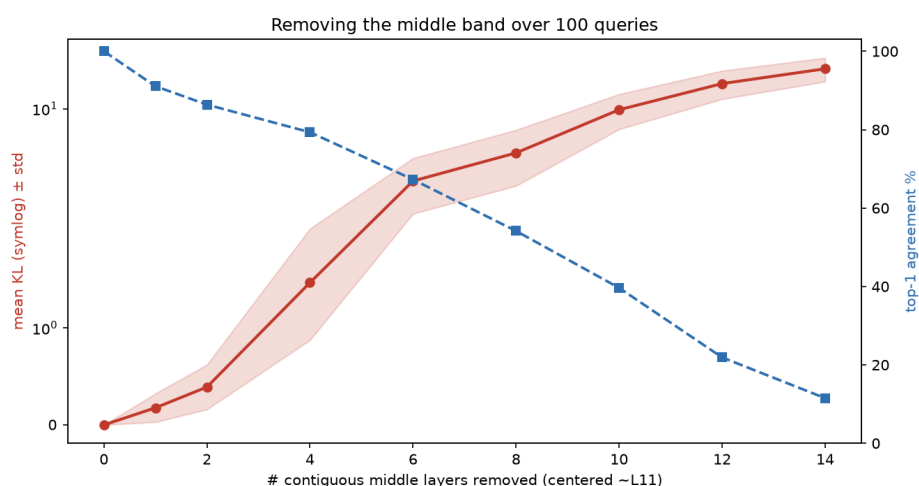
Single-layer sweep, now with variance bands — the early cliff is rock-solid:



Single-layer ablation over 100 queries

L0 → KL 17.2 / top-1 7%; L1 → KL 12.6 / top-1 24%; **every layer L2–L29 sits at top-1 89–94% (KL ≈ 0.1)**; only the final layer creeps up (L31: KL 1.18 / top-1 81%). Across 100 queries, exactly two layers are individually load-bearing.

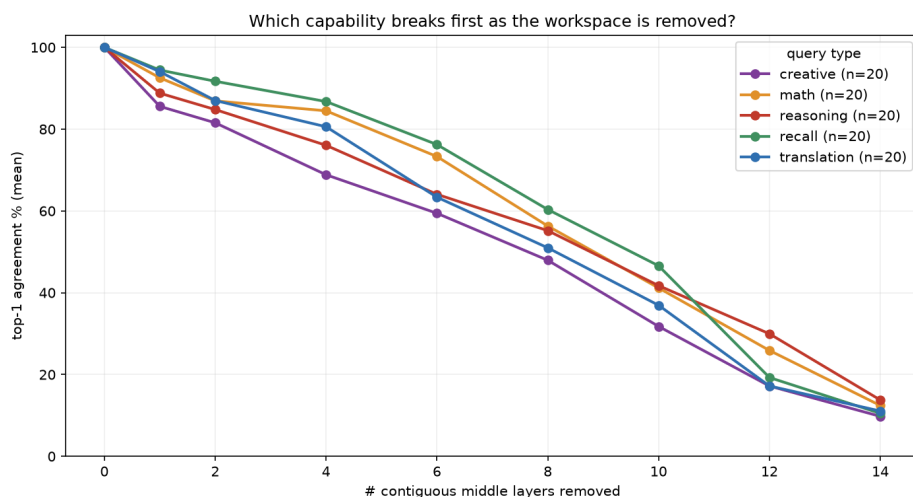
Cumulative band removal — the same graceful-then-catastrophic collapse, averaged over 100:



Cumulative removal over 100 queries

layers removed	1	2	4	6	8	10	12	14
KL	0.18	0.39	1.5	3.9	5.6	9.9	13.9	16.9
top-1	91%	86%	79%	67%	54%	40%	22%	12%

Which capability breaks first? Splitting the band-removal effect by query type:



Which capability breaks first, by type

Through the informative mid-range (4–10 layers removed) the ordering is consistent:

layers removed	recall	math	reasoning	translation	creative
4	87%	84%	76%	81%	69%
6	76%	73%	64%	63%	59%

layers removed	recall	math	reasoning	translation	creative
8	60%	56%	55%	51%	48%

Single-step recall is the most robust; open-ended / creative generation collapses first, with reasoning and translation in between. That’s the direction Anthropic’s workspace result predicts — abstract, generative capability leans hardest on the middle band, shallow recall leans on it least. Two honest caveats: the separation is **graded, not on/off**, and beyond ~12 layers removed *everything* converges toward collapse (the ordering gets noisy when the model is already broken). The metric is top-1 agreement with the intact model’s own greedy continuation — a proxy for “output preserved,” not a correctness grade.

7. The connection: a blunt version of the “global workspace”

Anthropic’s “**Verbalizable Representations Form a Global Workspace in Language Models**” (Gurnee, Sofroniew, ... Lindsey; Anthropic, July 2026) [10] identifies a **J-space** — a small set of token-aligned directions in intermediate layers (~6–10% of variance) that behaves like a cognitive-science *global workspace* [9]: a shared blackboard the model reads concepts from and reasons over. When they **suppress the top J-space directions**:

Impaired	Preserved
multi-hop reasoning (→ ~0%)	text parsing, grammatical fluency
creative / abstract generation	shallow classification, fact extraction
experiential self-report	single-step factual recall

Our coarse experiment — deleting whole *layers* rather than surgical *directions* — reproduces the **same asymmetry**: kill the middle band and reasoning/abstraction dies while shallow recall survives. And §3 shows *why* that band is special: it’s where topic is maximally **entangled yet fully decodable** — an information-rich, low-variance workspace, not a bottleneck.

Honest caveats. - We remove **entire layers** (sledgehammer); the J-space is a **low-rank subspace within** layers (scalpel). Layer ablation is a coarse, correlational proxy for workspace ablation. - Regime boundaries are eyeballed; §3’s decodability-vs-silhouette gap is the real evidence. - One model, 4-bit, teacher-forced KL over short greedy continuations, 400 texts /

8 prompts. Directional, not a benchmark. (Mean-pooling + UMAP also shape the *early*-layer picture; the invariant, method-robust claim is the decodable-but-entangled middle.)

Still — a satisfying result on a single T4: the layers where classes *stop looking separated* are precisely the layers you *can't* remove as a group without losing the ability to reason. That's where the workspace lives.

8. Reproduce

```
# On the GPU box (VM):
python server.py --load-4bit                # FastAPI on :8000
python experiments/run_experiments.py      # UMAP grid,
    separability, ablation (8-prompt)
python experiments/run_queries100.py      # 100-query sweep +
    per-category + CSV/JSON

# From your laptop: tunnel + open ui.html, ablate any set of
    layers interactively
gcloud compute ssh <vm> -- -N -L 8000:localhost:8000
#   ui.html → 0–31 toggle grid (presets: Concept L6–L17, Early,
    Late) → Generate & compare

python client.py "Explain why the sky is blue." --sweep 0 31 #
    CLI layer sweep → plot
```

9. References

1. nostalgebraist (2020). *Interpreting GPT: the Logit Lens*. LessWrong.
2. Belrose et al. (2023). *Eliciting Latent Predictions from Transformers with the Tuned Lens*. arXiv:2303.08112.
3. Meng, Bau, Andonian, Belinkov (2022). *Locating and Editing Factual Associations in GPT (ROME / causal tracing)*. NeurIPS. arXiv:2202.05262.
4. Tenney, Das, Pavlick (2019). *BERT Rediscovered the Classical NLP Pipeline*. ACL. arXiv:1905.05950.
5. Elhage et al. (2022). *Toy Models of Superposition*. Anthropic / Transformer Circuits.
6. Gromov, Tirumala, Shapourian, Glorioso, Roberts (2024). *The Unreasonable Ineffectiveness of the Deeper Layers*. arXiv:2403.17887.
7. Men et al. (2024). *ShortGPT: Layers in Large Language Models are More Redundant Than You Expect*. arXiv:2403.03853.
8. McInnes, Healy, Melville (2018). *UMAP*. arXiv:1802.03426. / van der Maaten & Hinton (2008). *t-SNE*. JMLR.
9. Baars (1988); Dehaene et al. — *Global Workspace Theory of consciousness (the analogy)*.

10. Gurnee, Sofroniew, Pearce, ... Lindsey (2026). **Verbalizable Representations Form a Global Workspace in Language Models.** Anthropic. <https://transformer-circuits.pub/2026/workspace/index.html>
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Built on a GCloud T4 VM · Llama-3.1-8B-Instruct (4-bit) · figures from experiments/run_experiments.py and figures/make_figure.py · numbers in blog/figures/metrics.json.